

Template for Computer Programming I Practical Recording

Date : 06/02/2025

Problem Specification : The problem is attempting to validate the date

Assumption : user input is valid.

Limitation : N/A

Input : Enter the day, month, and year of the date.

Processing : To check the year is leap year or not and the validity of the day, month, and year based on their conditions.

Output : Show whether the entered date is valid or not, and indicate if the year is a leap year.

Algorithm :

- Step1: prompt "Enter the month"
- Step2: Read the month given by user
- Step3: Assign input value for month
- Step4: Prompt "Enter the year"
- Step5: Read the year given by user
- Step6: Assign input value for year
- Step7: Prompt "Enter the day"
- Step8: Read the day given by user
- Step9: Assign input value for day
- Step10: Initialize a Boolean variable **monthValid** to check if the month is within the valid range ($0 \leq \text{month} \leq 12$)
- Step11: Initialize a Boolean variable **yearValid** to check if the year is within the valid range ($1000 \leq \text{year} \leq 1999$)
- Step12: Initialize a Boolean variable **leapYear** to check if the year is divisible by 400, or if it's divisible by 4 but not divisible by 100.
- Step13: Initialize daysInMonth as daysInMonth ← 0
- Step14: if(monthValid) is true

the program uses a switch statement to determine how many days are in that month. February is handled separately based on whether it's a leap year. (29 day in leap year otherwise 28 days).

month	days in month
1,3,5,7,8,12	31
4,6,9,10,11	30

Step15: Initialize a Boolean variable dayValid to check if the day is within the valid range ($1 \leq \text{day} \leq \text{daysInMonth}$)

Step16: if(monthValid && yearValid && dayValid) is true

Print as "Date is valid" then

if(leyear) is true then print as it's a leap year. Then go to step 18

Step17: Else print as "Date not is valid"

Step18: End the program

Programme listing : Programme file **attached**

Test data and expected output : 1. Test data: month : 2
Year : 1600
Day : 29

Expected output: Date is valid
it's a leap year

2. Test data: month : 2
Year : 1700
Day : 28

Expected output: Date is valid

3. Test data: month : 2
Year : 1605
Day : 29

Expected output: Date is not valid

Output obtained for test data : 1. Test data: month : 2
Year : 1700
Day : 29

Obtained output: Date is valid
it's a leap year

2. Test data: month : 2
 Year : 1700
 Day : 28

Obtained output: Date is valid

3. Test data: month : 2
 Year : 1605
 Day : 29

Obtained output: Date is not valid

Discussion : Take note of how to use Boolean values.

Conclusion : The user inputs the day, month, and year. The program then processes the input, checks the date's validity, and indicates if the year is a leap year.